

1. (a) Discuss how the creation and use of computer viruses is prohibited under the Computer Misuse Act.

[3]

Stating the computer misuse act does not prohibit act of creating of viruses [1]

Stating viruses can be used on equipment / data under the computer misuse act if the owners permission is first obtained [1]

Stating the creation / use of viruses is only prohibited if the virus is created / used to access or modify data or impair a computer without authorisation to do so [1]

- (b) A company has applied to copyright a database they have created. Their database contains a simple ordering of 10,000 novels alphabetically by title.

Discuss whether the company is likely to succeed in their attempt to copyright the database described above.

[4]

[1 mark] for stating that the company is unlikely to gain copyright protection for the database as described

[1 mark] each for points justifying / explaining why, e.g:

- Stating there must be a substantial investment in obtaining, verifying, or presenting the contents of the databased (which the presented one does not satisfy) [1]
- Stating the contents of the database must be original and the selection/arrangement of the database constitutes the author's own intellectual creation [1 mark]
- Giving an example of why the ordering is not sufficiently complex (e.g. ordering alphabetically does not have sufficient complexity or depth to be copyright in contrast to a more depth ordering such as a unique clustering by novel themes over time in different genres) [1 mark]

(c) A researcher is planning to conduct an experiment to investigate if a chatbot they have developed helps novice Java programmers solve lab exercises during a programming course.

The researcher plans to conduct the experiment on a Java Programming course they will teach next year during the mandatory tutorial sessions within the course.

Identify two potential ethical concerns with this research.

[4]

Looking for students to think about potential ethical issues with the proposed experiment. Any sensible answers are fine.

Give [1 each] for identifying an issue, with an additional [1] for a well justified point.

Good answers should mention some discussion of informed consent (which was repeatedly emphasised during the course):

- e.g., students might feel pressured into taking part in the research as the researcher teaches the class and pushes participation heavily on students.
- e.g., students might be unable to give consent (informed or not) since the experiment is interwoven with an essential teaching activity.

(d) You have been tasked with creating a survey to investigate consumers' frustration using a new virtual reality application, "MBP VR".

Describe one closed and one open-ended question that you could use in the survey to capture this.

[2]

[0.5] for any (relevant) closed question, [0.5] if good/well formatted:

- e.g., rate on a scale of 1-5 [1=not at all, 5=extremely frustrating] to what extent you were frustrated using MBP VR

[0.5] for any (relevant) open-ended question, [0.5] if a good/well formatted:

- e.g., can you describe, if at all, at which moments you were frustrated while using MBP VR and why

2. Locomotion systems in VR games provide ways for a player to navigate around a virtual environment presented by the VR system.

You have designed a new locomotion technique for a VR game you have made.

You believe your new locomotion system (“LocoPro”) will provide a better player experience by being easier to control and more immersive experience than the current locomotion system (“LocoOriginal”).

You plan to conduct a research project with an experiment (with human participants) to investigate this belief.

(a) Propose a hypothesis for an experiment to investigate this belief.

[2]

Correctly stated research question or hypothesis [1]

Give extra [1] if the hypothesis clearly indicates a well-chosen dependent variable

- *E.g., The LocoPro system will provide a more immersive experience than LocoOriginal [2]*

(b) Propose an experiment to investigate this hypothesis.

You should cover all aspects of a research project, except performing a literature survey and writing a research proposal.

[19]

I expect solutions to be wide ranging but should cover the key points outlined above in the question. There are more marks on offer here than the maximum awarded for the question, so students do not need to include everything to score well.

General guidelines for the breakdown of marks is as follows:

- Assumptions / constraints – max 3 marks
- Experimental design – max 12 marks
- Analysis methodology – max 3 marks
- Ethics (part of other) – max 3 marks
- Conclusion (part of other) – max 2 marks

Details on marks for each subsection follows:

- Appropriate list of reasonable assumptions or constraints [3]:
 - *e.g., participants are not experienced with LocoOriginal prior to conducting the experiment*
 - *e.g., participants are recruited with a variety of prior experience with VR and a mixture of genders and ages*
 - *e.g., experiment takes place in a controlled, lab environment to ensure there is sufficient empty space / that participants aren't at risk of tripping over while doing the experiment / that there is sufficient space for the locomotion system to function correctly*
- Experimental design:
 - Independent variables/conditions [1]: *e.g., locomotion method, LocoPro vs LocoOriginal*

- Control variables or describing decisions to reduce confounding factors [2]:
e.g., same headset/hardware used to evaluate each design, the same physical space to test/use the locomotion system in, etc.
- Dependent variables [2 max]: any good measures of perceived control or immersion in a VR context, e.g.,:
 - *task completion time (if a navigation task)*
 - *tracked in-game performance metrics (e.g. time taken, score achieved, etc)*
 - *standardised questionnaires to measure control/immersion*
 - *subjective ratings on perceived control,*
 - *subjective ratings on perceived immersion,*
 - *etc.*
- Type of experiment [1]: *e.g., within-subjects, between-subjects both fine.*
- Experiment tasks [2]: i.e., indicating what participants will be asked to do during the experiment period (*e.g., perform a navigation task around a maze, play a section of a game using both approaches*).
- Data collection method [1]: *e.g., tracked in-game metrics (e.g. time taken, performance scores), self-reported data, questionnaires, interviews with researchers, etc*
- Describing the form of the data collected [2].
- Pilot study [1]: i.e., testing data collection and processing in advance.
- Analysis method to be used [max 3 from the given details], must be specific with the terminology – ‘statistical methods’ or ‘statistical test’ is not sufficient; instead, expecting appropriate terms like: *independent/repeated-measures, two conditions, t-test, reference to null-hypothesis significance testing, test for ‘significant difference’, specifying the desired p-value for accepting the hypothesis, stating the null- and alternative-hypothesis*. If describing qualitative data an appropriate method such as open-axial-coding of interview statements or any other appropriate qualitative analysis technique.
- Ethics approval:
 - Stating a need to get ethics approval [1].
 - Mentioning obtaining informed consent from participants [1].
 - Mentioning a potential risk of the experiment to warn participants about (*e.g. participants could become motion sick during the experiment*) [1].
- Indicating how conclusions may be reached from the experiment – i.e., whether using the new locomotion system leads to improved [1].